

A Geometric and Computational Study of the Lonely Runner Conjecture for Speed Sets with

$$v_1 = 1 \text{ and } v_n \leq 2n$$

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Abstract

The Lonely Runner Conjecture asserts that for any set of n distinct positive integer speeds there exists a time at which every runner is at distance at least $1/(n+1)$ from a stationary observer on a circular track of unit circumference. The conjecture is known to hold for speed sets whose largest speed is linearly bounded in n , and Tao asked whether the bound $v_n \leq 2n$ can be reached. We observe that, together with the classical result $v_n \leq nv_1 \Rightarrow \kappa(V) \geq 1/(n+1)$, this question reduces entirely to the case $v_1 = 1$.

This report surveys the special cases and proof techniques relevant to that reduction, and describes the software developed during the project: an interactive visualiser that renders each runner's distance function as a piecewise-linear sawtooth and identifies the times realising the largest simultaneous separation, together with a family of exhaustive search programs that enumerate all admissible speed sets for a given n using exact rational arithmetic. We prove that the search over candidate times may be restricted to a finite set of breakpoints, and that whenever the maximiser is a crossing of two runners it has the form $\ell/(v_i + v_j)$, recovering geometrically the modular characterisation of κ due to Haralambis.

Using these tools we reverify the conjecture for all 125,476 speed sets with $v_1 = 1$ and $v_n \leq 2n$ for $n \leq 10$, and we test two candidate geometric properties that would, if true, provide a route to the case $v_1 = 1$. Both fail systematically throughout the searched range, the first as early as $n = 3$. *No counterexample to the Lonely Runner Conjecture itself is claimed or possible in this range*; each set admits a lonely time, but that time is not witnessed by the geometric feature the hypothesis predicted. We further show that six of seven natural descriptions of the speed pair realising κ fail exhaustively, while the seventh survives to $n = 7$ and then fails by a single equality, which constrains the shape any successful geometric argument can take. We observe that the bound is tight on the arithmetic progression $\{1, \dots, n\}$ for every n , though not only there: other tight sets exist, the smallest being $\{1, 3, 4, 7\}$. Both refuted hypotheses do hold on the arithmetic progression, which is likely why they seemed plausible, but that case turns out not to be representative of the general one.

Keywords: Lonely Runner Conjecture, Diophantine approximation, view obstruction, computational geometry, exhaustive search.

Contents

1	Introduction	4
1.1	The problem	4
1.2	Contributions	4
1.3	Attribution	5
1.4	Organisation	5
2	Preliminaries and Notation	5
2.1	The distance to the nearest integer	5
2.2	The loneliness constant	5
2.3	Elementary properties	6
3	Known Results	6
3.1	The gap of loneliness	6
3.2	Lacunary speed sets	6
3.3	Speeds of comparable size	7
3.4	Linearly bounded speeds	7
3.5	Additive structure	7
3.6	Verification for small numbers of runners	8
4	Reduction of the Case $v_n \leq 2n$ to $v_1 = 1$	8
5	The Geometric Framework	8
5.1	Sawtooth representation	8
5.2	Reduction to a finite search	9
5.3	The form of the candidate times	10
5.4	Valid lonely intersections	11
6	Software I: The Interactive Visualiser	12
6.1	Overview	12
6.2	Features	12
6.3	Implementation notes	13
7	Software II: Exhaustive Search	13
7.1	Enumeration	13
7.2	Analysis of a single set	14
7.3	Exact arithmetic and correctness	14
7.4	Validation	14
7.5	Complexity	15
7.6	Exhaustive verification results	15
8	Candidate Geometric Properties and Counterexamples	15
8.1	Motivation	15
8.2	Hypothesis H1 and its refutation	16
8.3	Hypothesis H2 and its refutation	16
8.4	Summary of counterexamples	18

9	Discussion	18
9.1	Why the hypotheses failed	18
9.2	The witnessing pairs admit no simple description	19
9.3	Limitations	20
10	Conclusion	20
11	Directions for Further Work	21
	Acknowledgements	21
A	Reproducing the Results	23

1 Introduction

1.1 The problem

Consider $n + 1$ runners on a circular track of unit circumference. All start at the same point at time $t = 0$ and thereafter run at pairwise distinct constant speeds. A runner is called *lonely* at time t if the distance along the track from that runner to every other runner is at least $1/(n + 1)$.

Conjecture 1.1 (Lonely Runner Conjecture). *For every $n \geq 1$ and every choice of $n + 1$ distinct speeds, each runner is lonely at some time.*

The conjecture originates in work of Wills [21] on inhomogeneous Diophantine approximation and was independently formulated by Cusick [8] in the geometric language of view obstruction; the name is due to Goddyn. A comprehensive account of its history and of the partial results known to date is given in the survey of Perarnau and Serra [14].

Two standard reductions simplify Conjecture 1.1 considerably. First, it suffices to fix attention on a single runner and place that runner at speed 0, so that the remaining n speeds are measured relative to it. Second, it suffices to consider speeds that are distinct positive integers. After these reductions the conjecture takes the following form, which we use throughout.

1.2 Contributions

The contributions of this report are:

- (C1) A short observation (Proposition 4.1) that Tao’s question of whether $v_n \leq 2n$ implies the conjecture reduces *entirely* to the case $v_1 = 1$, since the case $v_1 \geq 2$ follows immediately from the classical bound $v_n \leq nv_1$.
- (C2) A proof (Lemma 5.1) that the supremum defining $\kappa(V)$ is attained on an explicitly enumerable finite set of times, which justifies exhaustive search, together with the observation (Lemma 5.2 and Corollary 5.3) that the time realising $\kappa(V)$ always has the form $\ell/(v_i + v_j)$.
- (C3) An interactive visualisation tool for the piecewise-linear distance functions, including automatic identification of valid lonely intersections (Section 6).
- (C4) An exhaustive search program over all admissible speed sets, using exact rational arithmetic (Section 7).
- (C5) An independent exhaustive reverification of Conjecture 1.1 for all speed sets with $v_1 = 1$ and $v_n \leq 2n$ for $1 \leq n \leq 10$ (Section 7.6), serving as a validation of the implementation.
- (C6) Explicit counterexamples to two candidate geometric properties of the case $v_1 = 1$ (Section 8), each verified by hand, together with exhaustive counts of how often each property fails.
- (C7) Exhaustive testing of seven natural candidate properties of the witnessing pairs (Section 9), of which six fail outright and one survives to $n = 7$, narrowing the space of plausible geometric routes to the case $v_1 = 1$.

1.3 Attribution

This report is written as a survey of an ongoing investigation, and the majority of the mathematics it presents is not the author's own. Section 2 and Section 3 are entirely expository: every definition, theorem and proof technique in those sections is due to the cited authors, and is included to establish the context in which the case $v_1 = 1$ arises and to record the techniques studied during the research term. In particular, Theorem 3.1, Theorem 3.2, Theorem 3.3, Theorem 3.4 and Theorem 3.5 are quoted results, and the proof sketches given for them follow the cited sources and the survey of Perarnau and Serra [14].

What is original to this project is the following, and nothing beyond it: the observation of Proposition 4.1 (an immediate consequence of the known Theorem 3.3); the elementary structural results of Section 5; the software described in Section 6 and Section 7; and the computational results of Section 7.6, Section 8 and Section 9. No new case of Conjecture 1.1 is established.

1.4 Organisation

Section 2 fixes notation and records the elementary properties of κ . Section 3 surveys the known partial results that bear on the linearly bounded case. Section 4 carries out the reduction to $v_1 = 1$. Section 5 develops the geometric framework and proves the two lemmas underlying the search. Section 6 and Section 7 describe the software. Section 8 presents the hypotheses tested and the counterexamples found. Section 9 discusses what the data suggests, Section 10 summarises, and Section 11 lists open directions.

2 Preliminaries and Notation

2.1 The distance to the nearest integer

For $x \in \mathbb{R}$ write

$$\|x\| = \min\{x - \lfloor x \rfloor, \lceil x \rceil - x\} = \min_{k \in \mathbb{Z}} |x - k|,$$

the distance from x to the nearest integer. Thus $0 \leq \|x\| \leq 1/2$ for all x , and $\|x\| = 0$ exactly when $x \in \mathbb{Z}$. The function $x \mapsto \|x\|$ is periodic with period 1 and piecewise linear with slopes ± 1 .

If a runner has integer speed v and the stationary observer sits at the origin, then after time t the runner has travelled tv laps and the distance along the track between the runner and the observer is exactly $\|tv\|$.

2.2 The loneliness constant

Definition 2.1. For a finite set $V = \{v_1, v_2, \dots, v_n\}$ of distinct positive integers with $v_1 < v_2 < \dots < v_n$, define

$$\kappa(V) = \sup_{t \in (0,1)} \min_{1 \leq i \leq n} \|tv_i\|.$$

We refer to $f_V(t) = \min_i \|tv_i\|$ as the *lower envelope* of the speed set, and to a time t with $f_V(t) \geq 1/(n+1)$ as a *lonely time*. In this notation Conjecture 1.1 becomes the statement that

$$\kappa(V) \geq \frac{1}{n+1} \quad \text{for every such } V \text{ with } |V| = n.$$

Remark 2.2 (Notational convention). Throughout this report n denotes the number of *moving* runners, i.e. $n = |V|$, and the total number of runners is $n + 1$. The relevant bound is therefore

always $1/(n+1)$. This convention is used without exception, including in all tables and in the software.

2.3 Elementary properties

The following two facts are used repeatedly and are immediate from Definition 2.1.

Lemma 2.3 (Scaling invariance). *For every positive integer d , $\kappa(dV) = \kappa(V)$, where $dV = \{dv : v \in V\}$.*

Lemma 2.4 (Monotonicity). *If $V_1 \subseteq V_2$ then $\kappa(V_1) \geq \kappa(V_2)$.*

Proof. For Lemma 2.3, note that as t ranges over $(0, 1)$ the quantity $s = dt \bmod 1$ ranges over the whole of $(0, 1)$, and $\|t dv\| = \|sv\|$ for every v . The two suprema are therefore taken over the same set of values. For Lemma 2.4, the minimum in Definition 2.1 is taken over a larger index set for V_2 , hence is pointwise no larger, and the same holds after taking suprema. $\square$

Lemma 2.4 is the reason results for structured supersets (for instance arithmetic progressions) transfer to their subsets.

3 Known Results

This section collects the partial results relevant to the linearly bounded case. Proofs are sketched only to the extent needed later; full arguments may be found in the cited sources and in the survey [14].

3.1 The gap of loneliness

The earliest general lower bound is due to Wills.

Theorem 3.1 (Wills). *For every speed set V with $|V| = n$, $\kappa(V) \geq \frac{1}{2n}$.*

The proof is a union bound. For each i let $A_i = \{t \in (0, 1) : \|tv_i\| < 1/(2n)\}$ be the set of *bad times* for runner i ; each A_i has measure $1/n$, so the A_i cannot cover $(0, 1)$ unless they are pairwise disjoint, which cannot occur. Refinements of this argument, replacing the union bound by Bonferroni-type inequalities that account for the overlaps $A_i \cap A_j$, give the small improvements of Chen [6] and Chen–Cusick [7]; the best general lower bound currently known remains of the form $\frac{1}{2n} + o(1/n)$, most recently improved by Bedert [2]. The gap between this and the conjectured $\frac{1}{n+1}$ is the reason attention has shifted to structured families of speed sets, of which the linearly bounded case below is the one relevant here.

3.2 Lacunary speed sets

Theorem 3.2 (Pandey). *If $V = \{m_1 < m_2 < \dots < m_n\}$ satisfies*

$$\frac{m_{j+1}}{m_j} \cdot \frac{n-1}{n+1} \geq 2 \quad (1 \leq j \leq n-1),$$

then $\kappa(V) \geq \frac{1}{n+1}$.

The proof constructs a nested family of intervals $I_1 \supset I_2 \supset \cdots \supset I_n$ where every $t \in I_j$ is safe for runners $1, \dots, j$. The interval I_1 is the preimage under multiplication by m_1 of the safe region $[\frac{1}{n+1}, \frac{n}{n+1}]$, of length $\frac{1}{m_1} \cdot \frac{n-1}{n+1}$. The growth hypothesis guarantees that multiplying I_{j-1} by m_j stretches it to length at least 2, hence it contains a full unit interval $[\ell, \ell+1]$, and I_j is taken to be the preimage of the safe part of that lap. See [12].

3.3 Speeds of comparable size

The following classical result is the one on which Section 4 turns.

Theorem 3.3. *If $v_n \leq nv_1$ then $\kappa(V) \geq \frac{1}{n+1}$.*

Proof. Take $g = \frac{1}{v_1(n+1)}$. For each i we have $v_1 \leq v_i \leq nv_1$, so dividing through by $v_1(n+1)$,

$$\frac{1}{n+1} \leq gv_i = \frac{v_i}{v_1(n+1)} \leq \frac{n}{n+1}.$$

Every point of the interval $[\frac{1}{n+1}, \frac{n}{n+1}]$ is at distance at least $\frac{1}{n+1}$ from both 0 and 1, hence $\|gv_i\| \geq \frac{1}{n+1}$ for all i . $\square$

3.4 Linearly bounded speeds

Theorem 3.4 (Tao). *There is an absolute constant such that if $v_n < 1.2n$ then $\kappa(V) \geq \frac{1}{n+1}$.*

Tao's proof [19] is by contradiction: assuming $\kappa(V) < 1/(n+1)$ and testing the times $t = 1/j$ and nearby rationals forces a large collection of divisibility and membership conditions on V , ultimately forcing V to contain more than n elements. Tao asks explicitly whether the constant 1.2 can be replaced by 2, which would cover the arithmetic progression $\{1, 2, \dots, n\}$ and many other sets believed to be among the hardest instances.

Theorem 3.5 (Bohman–Peng). *For sufficiently large n , if $n < v_n \leq 2n - \exp(c(\log \log n)^2)$ then $\kappa(V) \geq \frac{1}{n+1}$.*

The method of [5] is that of *coprime mappings*: the possible speeds are partitioned into intervals, and a bijection between intervals is constructed in which every matched pair is coprime, using Hall's marriage theorem. Coprimality makes the matched runners behave independently modulo distinct integers, preventing them from clustering near the origin simultaneously. Pomerance [15] subsequently reduced the required interval length from $\exp(c(\log \log n)^2)$ to $c(\log n)^2$, bringing the admissible bound closer to $2n$; reaching $v_n \leq 2n$ exactly appears to require progress on the Jacobsthal problem.

3.5 Additive structure

Pandey [13] proves the conjecture for speed sets with small doubling. If $|V| = k \geq 4$ and $|V + V| = 2k - 1 + b < 3k - 3$, then by Freiman's $3k - 4$ theorem V is contained in an arithmetic progression of length at most $k + b$, and an explicit lower bound for progressions combined with Lemma 2.4 yields $\kappa(V) > 1/(k+1)$, provided that when the containing progression has common difference 1, its first term exceeds 1. The excluded case is exactly $a = 1$, which again singles out $v_1 = 1$ as the obstruction.

3.6 Verification for small numbers of runners

Conjecture 1.1 is known unconditionally for small numbers of runners. The literature states these results in terms of the total number of runners $k + 1$, where k is the number of nonzero speeds; this convention matches our n exactly, so a paper titled “ N lonely runners” settles the case $n = N - 1$.

n (moving)	total runners	Source
3	4	Betke and Wills [3], Cusick [8]
4	5	Cusick and Pomerance [9]
5	6	Bohman, Holzman and Kleitman [4]; shorter proof by Renault [16]
6	7	Barajas and Serra [1]
7	8	Rosenfeld [17]
8, 9	9, 10	Trakulthongchai [20]
10, 11, 12	11, 12, 13	Sungkawichai and Trakulthongchai [18]

Consequently Conjecture 1.1 holds for every $n \leq 12$. This is the fact that licenses the following statement, which we make once and rely on throughout: *no speed set considered anywhere in this report can be a counterexample to the Lonely Runner Conjecture*, since all of our searches lie in the range $n \leq 10$. Every “counterexample” reported below is a counterexample to a geometric hypothesis of our own formulation, and to nothing else.

These verifications rest on a reduction due to Tao [19]: if the conjecture holds for n speeds, then for $n + 1$ speeds it can fail only for a counterexample whose speeds are bounded by an explicit function of n , making each case a finite computation. The size of that computation was subsequently reduced substantially by Malikiosis, Santos and Schymura [11].

4 Reduction of the Case $v_n \leq 2n$ to $v_1 = 1$

The following observation motivates the remainder of the report. It is elementary, but it isolates precisely the residual difficulty in Tao’s question.

Proposition 4.1. *Let $V = \{v_1 < \dots < v_n\}$ with $v_n \leq 2n$. If $v_1 \geq 2$ then $\kappa(V) \geq \frac{1}{n+1}$.*

Proof. If $v_1 \geq 2$ then $nv_1 \geq 2n \geq v_n$, so the hypothesis of Theorem 3.3 is satisfied and the conclusion follows, with explicit witness $t = \frac{1}{v_1(n+1)}$. $\square$

Corollary 4.2. *Tao’s question, whether $v_n \leq 2n$ implies $\kappa(V) \geq 1/(n+1)$, is equivalent to the same statement restricted to speed sets with $v_1 = 1$.*

Accordingly, for the remainder of this report we study speed sets of the form

$$V = \{1 = v_1 < v_2 < \dots < v_n \leq 2n\}.$$

Remark 4.3. Proposition 4.1 also explains the hypothesis $a > 1$ in Pandey’s small-doubling theorem (Section 3.5): the two exclusions have the same source.

5 The Geometric Framework

5.1 Sawtooth representation

Plotting time t on the horizontal axis and $\|tv\|$ on the vertical axis represents runner v as a sawtooth: v congruent isosceles triangles over $[0, 1]$, each of base $1/v$ and height $1/2$, with peaks at $t = \frac{2k+1}{2v}$

and zeros at $t = k/v$. Superimposing the sawtooths of all n runners, the quantity $f_V(t) = \min_i \|tv_i\|$ is the lower envelope of the resulting arrangement, and $\kappa(V)$ is the height of the highest point of that envelope. The conjecture asserts that the envelope rises to height at least $1/(n+1)$ somewhere.

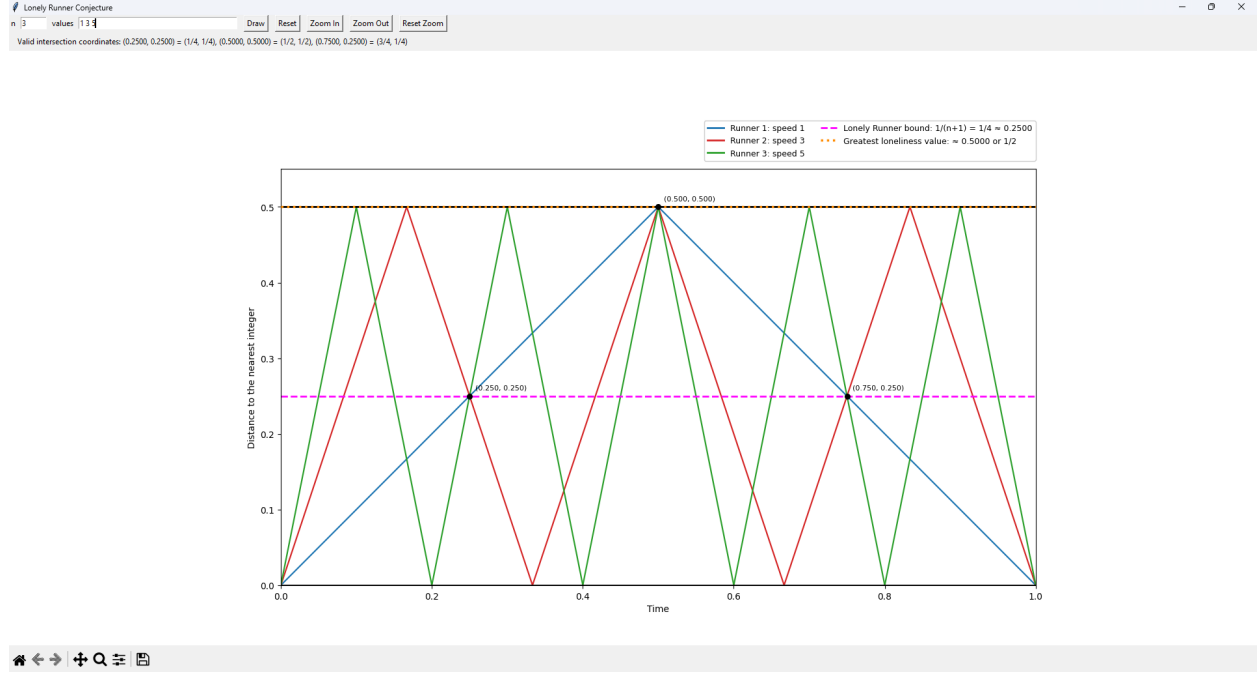


Figure 1: The visualiser, `LRCgraph.py`, on $V = \{1, 3, 5\}$ ($n = 3$). Each runner's distance function $\|tv_i\|$ is drawn as a sawtooth in its own colour and identified in the legend by index and speed. The dashed line marks the bound $1/(n+1) = 1/4$; the dotted line marks the computed value $\kappa(V) = 1/2$. Black points are the valid lonely intersections of Definition 5.6, with exact coordinates listed above the plot: $(1/4, 1/4)$, $(1/2, 1/2)$ and $(3/4, 1/4)$. Zoom and cursor-readout controls are provided for examining dense regions at larger n .

The set $\{1, 3, 5\}$ in Figure 1 illustrates the extreme case in which every speed is odd. At $t = 1/2$ we have $\|v/2\| = 1/2$ for every odd v , so all three sawtooths peak simultaneously and $\kappa(V) = 1/2$, the largest value the function can take. More generally $\kappa(V) = 1/2$ whenever every speed in V is odd. The figure also shows why Remark 5.8 is rarely binding in practice: when a peak does realise the maximum, the peaks of several runners typically coincide there, so the point is registered as a crossing after all.

5.2 Reduction to a finite search

The supremum in Definition 2.1 is over a continuum, but it is attained on a finite set that can be written down explicitly.

Lemma 5.1 (Finite search). *The function f_V is continuous and piecewise linear on $[0, 1]$, and its breakpoints lie in the finite set*

$$B(V) = \underbrace{\left\{ \frac{2k+1}{2v_i} \right\}}_{\text{peaks}} \cup \underbrace{\left\{ t : \|tv_i\| = \|tv_j\|, i \neq j \right\}}_{\text{crossings}} \cap (0, 1).$$

Consequently $\kappa(V) = \max_{t \in B(V)} f_V(t)$, and $\kappa(V)$ is rational.

Proof. Fix i . The function $t \mapsto \|tv_i\|$ is piecewise linear on $[0, 1]$ with slopes $\pm v_i$, changing slope only at the zeros $t = k/v_i$ (from $-v_i$ to $+v_i$) and at the peaks $t = (2k+1)/(2v_i)$ (from $+v_i$ to $-v_i$). Each is therefore piecewise linear with finitely many pieces, and so is their pointwise minimum f_V , which is moreover continuous as a minimum of finitely many continuous functions.

Let $t_0 \in (0, 1)$ be a point at which f_V changes slope, and let

$$A(t_0) = \{i : \|t_0 v_i\| = f_V(t_0)\}$$

be the set of *active* runners there. We distinguish two cases.

If $A(t_0) = \{i\}$ is a single index, then by continuity runner i remains strictly below every other runner on some neighbourhood of t_0 , so f_V agrees with $\|tv_i\|$ there. A slope change of f_V is then a slope change of $\|tv_i\|$, hence t_0 is a zero or a peak of runner i .

Otherwise $|A(t_0)| \geq 2$, so there are indices $i \neq j$ with $\|t_0 v_i\| = \|t_0 v_j\| = f_V(t_0)$, which is a crossing. Hence every breakpoint of f_V lies in $B(V) \cup \{k/v_i\}$.

A continuous piecewise-linear function on a compact interval attains its maximum at a breakpoint or an endpoint. The endpoints give $f_V(0) = f_V(1) = 0$, and each zero $t = k/v_i$ gives $f_V(t) = 0$ as well, so none of these can be the maximum unless $\kappa(V) = 0$, which is impossible since $\kappa(V) > 0$ for any nonempty V . The zeros may therefore be discarded, and the maximum is attained on $B(V)$.

Finally, every element of $B(V)$ is rational: peaks are rational by inspection, and a crossing is the solution of a linear equation with integer coefficients (Lemma 5.2). Since f_V maps rationals to rationals, $\kappa(V)$ is rational. $\square$

Lemma 5.1 is what makes the exhaustive search of Section 7 legitimate rather than merely heuristic. It is also the reason every computation can be carried out in exact rational arithmetic, since the finite set of candidate times consists entirely of rationals.

5.3 The form of the candidate times

Lemma 5.2 (Crossing times). *Let the graphs of distinct runners v_i and v_j cross at $t \in (0, 1)$.*

(i) *If one edge is rising and the other falling, then $t = \frac{\ell}{v_i + v_j}$ for some integer ℓ .*

(ii) *If both edges rise, or both fall, then $t = \frac{\ell}{v_i - v_j}$ for some integer ℓ .*

Proof. A rising edge of runner i has equation $y = v_i t - k$ and a falling edge $y = (k+1) - v_i t$. In case (i), equating $v_i t - k$ with $(k'+1) - v_j t$ gives $t(v_i + v_j) = k + k' + 1$. In case (ii), equating $v_i t - k$ with $v_j t - k'$ gives $t(v_i - v_j) = k - k'$; the falling–falling case is identical up to sign. $\square$

Case (ii) crossings are never maximisers, which is what allows us to ignore them when computing κ .

Corollary 5.3. $\kappa(V)$ is attained either at a peak $t = \frac{2k+1}{2v_i}$ or at a rising–falling crossing; in the latter case $\kappa(V) = \frac{\ell'}{v_i + v_j}$ for some integers ℓ', i, j .

Proof. By Lemma 5.1 the maximum lies at a breakpoint, and zeros are excluded there, so let t be a breakpoint at which $\kappa(V)$ is attained and let $A(t)$ be the set of active runners as in the proof of Lemma 5.1. By continuity the inactive runners remain strictly above on a neighbourhood of t , so f_V agrees with $\min_{i \in A(t)} \|tv_i\|$ there.

If some active runner has a peak at t , we are in the first case. Otherwise every active runner's graph passes through $(t, f_V(t))$ along a single rising or falling edge. Were all these edges rising, f_V would be strictly increasing on a neighbourhood of t , and were they all falling it would be strictly decreasing; in either case t would not be a maximum. Hence some active runner v_i is rising at t and some active runner v_j is falling. These are distinct, since a rising and a falling edge of the same runner meet only at a peak, which is excluded. Their graphs cross at t , and Lemma 5.2(i) gives $t = \ell'/(v_i + v_j)$. $\square$

Remark 5.4. The pair (v_i, v_j) produced by this argument need not be unique, for two separate reasons. First, three or more runners may be active at the maximum, in which case every rising-falling pair among them realises it: in $V = \{1, 2, 4, 5\}$ the value $\kappa(V) = 1/3$ is attained at $t = 1/3$ with all four runners active, and the pairs $(1, 2)$, $(1, 5)$, $(2, 4)$ and $(4, 5)$ all represent it. Second, the maximum may be attained at several distinct times: in $V = \{1, 4, 5, 6, 7\}$ the value $\kappa(V) = 2/11$ occurs both at $t = 4/11$, realised by $(5, 6)$, and at $t = 5/11$, realised by $(4, 7)$. Any statement about “the” realising pair must therefore be read as a statement about *some* such pair, and this is the reading used in Section 9.2.

Remark 5.5. Corollary 5.3 recovers geometrically the third of Haralambis' characterisations of κ [10], namely

$$\kappa(V) = \max_{m=v_j+v_l} \frac{1}{m} \min_i |xv_i|_m, \quad 1 \leq x \leq \frac{m}{2},$$

where $|x|_m$ denotes the absolute least residue of x modulo m ; this is the form used by Pandey in the proofs of Section 3.5. The index l is allowed to equal j there, and this is what accommodates the peak case of Corollary 5.3: a peak of runner v_i occurs at $t = (2k+1)/(2v_i) = (2k+1)/(v_i + v_i)$, so both cases of the corollary give a time of the form ℓ/m with m a sum of two (not necessarily distinct) speeds. The visualiser is therefore not an ad hoc construction: the times it marks as maximisers are exactly the times arising from the moduli $v_j + v_l$ that the modular approach already singles out. We regard this correspondence as the main conceptual justification for the geometric approach.

5.4 Valid lonely intersections

Definition 5.6. A point (t_0, y_0) is a *valid lonely intersection* for V if

- (i) t_0 is a crossing of the graphs of two distinct runners $v_i \neq v_j$ at height $y_0 = \|t_0 v_i\| = \|t_0 v_j\|$;
- (ii) $y_0 \geq \frac{1}{n+1}$; and
- (iii) $\|t_0 v\| \geq y_0$ for every $v \in V$, i.e. no runner lies strictly below the crossing.

We say the intersection *involves* the speeds v_i and v_j .

Condition (iii) says the crossing lies on the lower envelope rather than merely in the arrangement. Every valid lonely intersection is a lonely time, since (ii) and (iii) together give $f_V(t_0) \geq y_0 \geq 1/(n+1)$.

Two remarks are needed to relate Definition 5.6 to Corollary 5.3, as the two notions are close but not identical.

Remark 5.7 (Valid intersections need not be maximisers). Definition 5.6 admits crossings of type (ii) of Lemma 5.2. Such a crossing is a genuine lonely time, so its inclusion is harmless for hypotheses phrased as “some valid lonely intersection has property P ”, but it need not be a point at which the

envelope attains its maximum, and the first valid intersection reported by the software is in general *not* the maximiser. Note that a type-(ii) crossing may nonetheless *occur* at the maximum: in $V = \{1, 2, 4\}$ the runners 1 and 4 are both rising at $t = 1/3$ where $\kappa(V) = 1/3$ is attained, the maximum being realised there by the rising–falling pair $(1, 2)$. We verified computationally that restricting Definition 5.6 to type-(i) crossings changes none of the hypothesis verdicts reported in Section 8, for every n searched.

Remark 5.8 (Peak-attained maxima). Definition 5.6 excludes the case where $\kappa(V)$ is attained at a peak of a single runner and at no crossing. This requires every other runner to lie at height exactly $1/2$ simultaneously. Across all speed sets with $v_1 = 1$ and $v_n \leq 2n$ for $n \leq 6$, the only such set is the trivial $V = \{1\}$. The exclusion is therefore immaterial in the range searched, but a complete implementation should test peaks as well; see Section 11.

6 Software I: The Interactive Visualiser

6.1 Overview

Five programs were written during the project, all in Python 3 and all using `fractions.Fraction` for every arithmetic operation. The visualiser and the three interactive search tools use `tkinter`; `patterns.py` runs in a terminal. One program is a visualisation tool and the other four are search tools:

Program	Purpose	Section
<code>LRCgraph.py</code>	interactive visualiser	Section 6
<code>lrc.py</code>	verification of Conjecture 1.1, $v_1 = 1$, $v_n \leq 2n$	Section 7.6
<code>LRCTestable.py</code>	test of Hypothesis 1	Section 8
<code>LRCTestable2.py</code>	test of Hypothesis 2	Section 8
<code>patterns.py</code>	envelope maximisers and their speed pairs	Section 9

All five programs, together with the complete output underlying every table in this report, are available at <https://github.com/asfihassan/lonely-runner-geometric-search>.

The visualiser, `LRCgraph.py`, takes the number of runners n and a space-separated list of speeds, and renders the superimposed sawtooths with `matplotlib` embedded in the window via `FigureCanvasTkAgg`. The four search programs share the same mathematical core, namely the functions `height`, `event_times`, `graph_segments`, `segment_intersection` and `valid_intersections`. They differ only in which speed sets they enumerate and which property of the resulting intersections they test.

6.2 Features

- Rendering of each runner’s sawtooth in a distinct colour, with a legend identifying runner index and speed.
- A dashed horizontal reference line at $1/(n + 1)$, computed from the input.
- Detection of all pairwise segment crossings, filtered by Definition 5.6, and display of the surviving points with their exact coordinates.
- Computation of the greatest loneliness value $\kappa(V)$ over the valid points, displayed as both a decimal and a reduced fraction, with a second reference line when it exceeds $1/(n + 1)$.

- Interactive zoom (mouse wheel and buttons), cursor coordinate readout, and reset.

6.3 Implementation notes

Each runner's sawtooth is decomposed into oriented linear segments. Each segment is stored as a tuple recording its left and right endpoints, its slope and intercept, and the index and speed of the runner it belongs to. Runner v contributes $2v$ segments: a rising segment $y = vt - k$ on $[k/v, (2k+1)/2v]$ and a falling segment $y = (k+1) - vt$ on $[(2k+1)/2v, (k+1)/v]$, for $0 \leq k < v$. All endpoints, slopes and intercepts are exact rationals.

Two segments are intersected by solving the linear system directly; segments belonging to the same runner and parallel segments are rejected first, and the solution is retained only if it lies within both parameter ranges and inside $(0, 1)$. The envelope test of Definition 5.6(iii) is then applied by evaluating $\|t_0v\|$ for every $v \in V$.

```

1 def segment_intersection(segment1, segment2):
2     (left1, right1, slope1, intercept1, runner1, speed1) = segment1
3     (left2, right2, slope2, intercept2, runner2, speed2) = segment2
4
5     if runner1 == runner2:
6         return None
7     if slope1 == slope2:
8         return None
9
10    t = (intercept2 - intercept1) / (slope1 - slope2)
11
12    if not (left1 <= t <= right1 and left2 <= t <= right2
13            and Fraction(0) < t < Fraction(1)):
14        return None
15
16    return (t, slope1 * t + intercept1, runner1, runner2)

```

Listing 1: Exact segment intersection (excerpt).

The zoom controls turned out to matter more than expected. The number of segments grows as $2 \sum_i v_i$, so for $n = 7$ with $v_n \leq 14$ the plot can hold several hundred segments, at which scale the valid intersections are impossible to tell apart by eye. Magnification was needed to inspect the counterexamples of Section 8 visually before confirming them numerically.

7 Software II: Exhaustive Search

7.1 Enumeration

For a fixed n the program enumerates every strictly increasing speed set satisfying the prescribed constraints. Two modes are supported: the maximum speed may be required to equal the bound exactly, or merely to be at most the bound. The speeds 1 and 2 are placed in every set before enumeration when the hypothesis under test requires them, so that sets omitting them are never generated.

```

1 def enumerate_speed_sets(n, last_equals_bound):
2     upper = 2 * n - 3
3     if n < 3 or upper < 3:
4         return
5
6     if last_equals_bound:

```

```

7     for middle in itertools.combinations(range(3, upper), n - 3):
8         yield (1, 2) + middle + (upper,)
9     else:
10    for rest in itertools.combinations(range(3, upper + 1), n - 2):
11        yield (1, 2) + rest

```

Listing 2: Enumeration of admissible speed sets with $v_1 = 1$, $v_2 = 2$, $v_n \leq 2n - 3$.

7.2 Analysis of a single set

For each generated V the program computes the bound $1/(n+1)$, determines whether a lonely time exists, enumerates all valid lonely intersections in the sense of Definition 5.6, and records which speeds witness them. Each set is then classified as

- (a) satisfying the hypothesis under test;
- (b) a counterexample to the hypothesis (a lonely time exists, but no valid intersection has the required form); or
- (c) a counterexample to the Lonely Runner Conjecture itself (no lonely time exists).

No set of type (c) was found, and none can exist within the ranges searched, since the conjecture is verified for these values of n (Section 3.6). Category (c) is retained purely as a correctness check on the implementation.

7.3 Exact arithmetic and correctness

All computation is performed over $\mathbb{Q}$ using Python’s `fractions.Fraction`; no floating-point comparison occurs anywhere in the search, and consequently every quantity reported in this paper is exact.

The reason is not stylistic. The minimum of $\kappa(V) - \frac{1}{n+1}$ over all speed sets with $v_1 = 1$ and $v_n \leq 2n$ is *exactly zero* for every n in the searched range, and it is attained by the arithmetic progression

$$V = \{1, 2, \dots, n\}, \quad \kappa(V) = \frac{1}{n+1},$$

witnessed at $t = \frac{1}{n+1}$ by the pair $(v_1, v_n) = (1, n)$. The conjectured bound is therefore tight on the family Tao identifies as the hardest instance, and any tolerance-based comparison $y \geq \frac{1}{n+1} - \varepsilon$ or $y > \frac{1}{n+1}$ would misclassify it. Working over the rationals is a requirement for correctness rather than a matter of taste.

The arithmetic progression is not, however, the only tight set. Counting the speed sets that attain equality gives one for $n = 2, 3$ and 6 , two for $n = 4$ and 5 , and three for $n = 7$. The smallest example beyond the progression is $V = \{1, 3, 4, 7\}$, where $t = 2/5$ gives distances $\frac{2}{5}, \frac{1}{5}, \frac{2}{5}, \frac{1}{5}$ and hence $\kappa(V) = 1/5$ exactly. Whether the extremal sets admit a clean description is a natural question we have not pursued.

7.4 Validation

The implementation was validated in three ways: against the closed form $\kappa(\{v_1, v_2\}) = \lfloor (v_1 + v_2)/2 \rfloor / (v_1 + v_2)$, valid for two coprime runners (for $\gcd(v_1, v_2) = d > 1$ apply Lemma 2.3 first); against the tight value $\kappa(\{1, \dots, n\}) = 1/(n+1)$ above; and against the published verification of Conjecture 1.1 for small n (Section 3.6), which the search reproduces in full (Section 7.6).

7.5 Complexity

Two distinct computations are performed per speed set, with very different costs. Deciding whether a lonely time exists requires only the event-time scan, costing $O(n \sum_i v_i) = O(n^3)$ per set. Enumerating all valid intersections requires the pairwise segment test, costing $O(n(\sum_i v_i)^2) = O(n^5)$ per set. Since the number of admissible sets is $\binom{2n-1}{n-1}$, the second computation is the binding constraint, and it is the reason the hypothesis searches reach smaller n than the verification search.

7.6 Exhaustive verification results

Table 1 reports an exhaustive search over every speed set with $v_1 = 1$ and $v_n \leq 2n$, for $1 \leq n \leq 10$.

Table 1: Exhaustive verification of Conjecture 1.1 for $v_1 = 1$, $v_n \leq 2n$. Timings are for a single-threaded CPython 3.14.3 run on an Intel Core i5-12400F (6 cores, 12 threads, 2.50 GHz base) with 16 GB RAM.

n	speed sets $\binom{2n-1}{n-1}$	sets with no lonely time	time (s)
1	1	0	< 0.01
2	3	0	< 0.01
3	10	0	< 0.01
4	35	0	0.01
5	126	0	0.06
6	462	0	0.30
7	1 716	0	1.63
8	6 435	0	7.30
9	24 310	0	35.66
10	92 378	0	197.76
Total	125 476	0	242.72

No speed set in this range fails to admit a lonely time. We stress that this is a *reverification*: Conjecture 1.1 is already known unconditionally for $n \leq 12$ (Section 3.6), so no new case of the conjecture is established here. The value of the computation is as an independent check on the implementation. The route taken here, exact analysis of a piecewise-linear envelope, has nothing methodologically in common with the modular sieves used in the published verifications, so agreement between the two across 125 476 instances is meaningful evidence that the tools are correct. Those same tools are used in Section 8, where the results are not already known and no independent check is available.

8 Candidate Geometric Properties and Counterexamples

8.1 Motivation

By Proposition 4.1 the outstanding case is $v_1 = 1$. A natural strategy is to look for a geometric feature that is always present and always witnesses a lonely time; if such a feature could be identified and its presence proved, the case would follow. The runner of speed 1 is a natural candidate, since its graph is a single triangle spanning $[0, 1]$ and is therefore above $1/(n+1)$ on all but a short initial and final segment.

Lonely Runner — Exhaustive Speed-1-or-Speed-2 Intersection Search

n (runners): 5 last runner speed: $v_n \leq 2n-3$ all $v_i \leq 2n-3$ Run Clear

n = 5 mode: $v_n \leq 2n-3$ required speeds: $v_1 = 1$ and $v_n = 2$ cases: 10 | HYPOTHESIS FAILS IN 1 CASE(S).

Speeds	$1/(n+1)$	Lonely time exists?	First valid time t	Minimum height	Speed-1 or speed-2 intersection?	First speed-1/2 t	Intersection height	Intersecting speeds	Verdict
12345	1/6	yes	1/6	1/6	yes	1/6	1/6	1 and 5	Valid intersection involving the speed-1 runner or speed-2 runner found
12346	1/6	yes	1/5	1/5	yes	1/5	1/5	1 and 4	Valid intersection involving the speed-1 runner or speed-2 runner found
12347	1/6	yes	1/6	1/6	yes	1/6	1/6	1 and 7	Valid intersection involving the speed-1 runner or speed-2 runner found
12356	1/6	yes	1/4	1/4	yes	1/4	1/4	1 and 3	Valid intersection involving the speed-1 runner or speed-2 runner found
12357	1/6	yes	1/6	1/6	yes	1/6	1/6	1 and 5	Valid intersection involving the speed-1 runner or speed-2 runner found
12367	1/6	yes	1/5	1/5	yes	1/5	1/5	1 and 6	Valid intersection involving the speed-1 runner or speed-2 runner found
12456	1/6	yes	3/10	1/5	no				Lonely time exists, but no valid intersection involving the speed-1 runner or speed-2 runner found
12457	1/6	yes	1/6	1/6	yes	1/6	1/6	1 and 5	Valid intersection involving the speed-1 runner or speed-2 runner found
12467	1/6	yes	1/5	1/5	yes	1/5	1/5	1 and 4	Valid intersection involving the speed-1 runner or speed-2 runner found
12567	1/6	yes	1/4	1/4	yes	1/4	1/4	1 and 3	Valid intersection involving the speed-1 runner or speed-2 runner found

Figure 2: The exhaustive search program `LRctable2.py` testing Hypothesis 2 at $n = 5$ in the mode $v_n \leq 2n - 3 = 7$. All ten admissible speed sets containing both 1 and 2 are shown, with the bound $1/(n + 1)$, whether a lonely time exists, the first valid lonely intersection, the speeds realising it, and the classification. Nine sets satisfy the hypothesis (green); the single failure (red) is $V = \{1, 2, 4, 5, 6\}$ of Example 8.3, for which a lonely time exists at $t = 3/10$ with height $1/5$ but no valid intersection involves the runner of speed 1 or speed 2.

8.2 Hypothesis H1 and its refutation

Hypothesis 1. For every speed set V with $v_1 = 1$ and $v_n \leq 2n$, there exists a valid lonely intersection involving the runner of speed 1.

Hypothesis 1 is false, and fails already at $n = 3$.

Example 8.1 ($V = \{1, 3, 4\}$, the smallest counterexample). Here $n = 3$ and the bound is $1/4$. At $t = 3/7$ the runner distances are

$$\left\| \frac{3}{7} \right\| = \frac{3}{7}, \quad \left\| \frac{9}{7} \right\| = \frac{2}{7}, \quad \left\| \frac{12}{7} \right\| = \frac{2}{7},$$

so $f_V(3/7) = 2/7 > 1/4$ and $t = 3/7$ is a lonely time. The envelope maximum is attained by the runners of speeds 3 and 4, with $3/7 = 3/(3+4)$ as predicted by Lemma 5.2(i). The runner of speed 1 sits at $3/7$, strictly above the envelope. The set has 16 segments and 18 crossings in total, and exhaustive inspection of all of them confirms that none satisfying Definition 5.6 involves speed 1.

Since a lonely time does exist here, Example 8.1 refutes the geometric hypothesis alone and is entirely consistent with Conjecture 1.1. The failure is also not an isolated one. Table 2 shows that the proportion of speed sets violating Hypothesis 1 stays in the region of ten per cent and shows no sign of falling away as n grows.

The extremal mode $v_n = 2n$ is the one directly relevant to Tao’s question, and the hypothesis fails there too, first at $V = \{1, 3, 5, 8\}$ when $n = 4$. Failures are proportionally more common in the extremal mode than in the general one, reaching roughly one set in seven by $n = 6$, so restricting attention to the hardest instances does not make the geometric property more likely to hold.

8.3 Hypothesis H2 and its refutation

The natural weakening admits the runner of speed 2 as an alternative witness. The hypothesis below is stated for speed sets containing both 1 and 2, and this is the form the search program tests.

The restriction to sets containing 2 is not an exclusion of some untreated case. If $2 \notin V$ then there is no speed-2 runner, and the condition “some valid lonely intersection involves the runner of speed 1 or the runner of speed 2” says exactly what Hypothesis 1 says. Those sets were therefore already covered by the search of Table 2, which ranges over every set with $v_1 = 1$, and Hypothesis 1 fails on them: $V = \{1, 3, 4, 5\}$ has $n = 4$ and $v_n = 5 = 2n - 3$ with $2 \notin V$, and admits no valid lonely

Table 2: Failures of Hypothesis 1 over all speed sets with $v_1 = 1$, in both modes offered by `LRCtable.py`. Repeating the count with Definition 5.6 restricted to rising–falling crossings (Remark 5.7) gives identical figures throughout, confirming that the treatment of same-direction crossings does not affect the conclusion.

n	$v_n \leq 2n$		$v_n = 2n$	
	speed sets	counterexamples	speed sets	counterexamples
2	3	0	1	0
3	10	1	4	0
4	35	3	15	1
5	126	15	56	8
6	462	49	210	32
7	1 716	146	792	79

intersection involving the speed-1 runner. Restricting to sets containing both 1 and 2 is simply the only regime in which Hypothesis 2 differs from Hypothesis 1 at all.

Remark 8.2. One thing can be said about the sets with $2 \notin V$ that does not follow from Hypothesis 1. Suppose $v_1 = 1$, $2 \notin V$ and $v_n \leq 2n - 3$. Then $v_2 \geq 3$, so $V \setminus \{1\}$ has $n - 1$ elements with smallest at least 3 and largest at most $2n - 3$; since $3(n - 1) \geq 2n - 3$, Theorem 3.3 applies and gives $\kappa(V \setminus \{1\}) \geq 1/n$. The remaining $n - 1$ runners can thus be made simultaneously lonely with margin better than required. What this does not do is place the speed-1 runner, since the witness $t = 1/(v_2 n) \leq 1/(3n)$ leaves it too close to the origin. Whether the slack between $1/n$ and $1/(n + 1)$ can be used to find a better time is open.

Hypothesis 2. For every speed set V with $v_1 = 1$, $v_2 = 2$ and $v_n \leq 2n - 3$, there exists a valid lonely intersection involving the runner of speed 1 or the runner of speed 2.

Hypothesis 2 is also false, in both the strict and the extremal regime.

Example 8.3 ($V = \{1, 2, 4, 5, 6\}$, $v_n < 2n - 3$). Here $n = 5$, the bound is $1/6$, and $v_n = 6 < 7 = 2n - 3$. At $t = 3/10$,

$$\left\| \frac{3}{10} \right\| = \frac{3}{10}, \quad \left\| \frac{6}{10} \right\| = \frac{2}{5}, \quad \left\| \frac{12}{10} \right\| = \frac{1}{5}, \quad \left\| \frac{15}{10} \right\| = \frac{1}{2}, \quad \left\| \frac{18}{10} \right\| = \frac{1}{5},$$

so $f_V(3/10) = 1/5 > 1/6$. The envelope maximum is attained by the runners of speeds 4 and 6, with $3/10 = 3/(4 + 6)$. A second valid lonely intersection occurs at $t = 4/11$ at height $2/11$, involving speeds 5 and 6. Neither the runner of speed 1 nor that of speed 2 participates in any valid lonely intersection.

Example 8.4 ($V = \{1, 2, 5, 6, 7, 8, 11\}$, $v_n = 2n - 3$). Here $n = 7$, the bound is $1/8$, and $v_n = 11 = 2n - 3$. At $t = 3/13$ the distances are

$$\frac{3}{13}, \quad \frac{6}{13}, \quad \frac{2}{13}, \quad \frac{5}{13}, \quad \frac{5}{13}, \quad \frac{2}{13}, \quad \frac{6}{13}$$

for the speeds 1, 2, 5, 6, 7, 8, 11 respectively, so $f_V(3/13) = 2/13 > 1/8$. The minimum is attained by the runners of speeds 5 and 8, with $3/13 = 3/(5 + 8)$. Again no valid lonely intersection involves speed 1 or speed 2.

As with Hypothesis 1, the failures are systematic rather than sporadic. Table 3 gives the exhaustive counts in both modes. Note that Example 8.3 and Example 8.4 are each the *first* failure in their respective mode: below $n = 5$ (resp. $n = 7$) the hypothesis holds for every admissible set, which is presumably why it survived initial hand-testing.

Table 3: Failures of Hypothesis 2 over all speed sets containing both 1 and 2 with $v_n \leq 2n - 3$ (left) and $v_n = 2n - 3$ (right).

n	$v_n \leq 2n - 3$		$v_n = 2n - 3$	
	speed sets	counterexamples	speed sets	counterexamples
3	1	0	1	0
4	3	0	2	0
5	10	1	6	0
6	35	2	20	0
7	126	7	70	1
8	462	17	252	3
9	1 716	52	924	11

8.4 Summary of counterexamples

Table 4: Counterexamples to the geometric hypotheses. In every case a lonely time exists, so none of these sets contradicts Conjecture 1.1. The witness shown is the maximiser of the lower envelope; the modulus column illustrates Lemma 5.2.

Speed set V	n	$\frac{1}{n+1}$	witness t	$\kappa(V)$	witnessing pair	$m = v_i + v_j$	refutes
$\{1, 3, 4\}$	3	1/4	3/7	2/7	(3, 4)	7	H1
$\{1, 3, 4, 5\}$	4	1/5	4/9	2/9	(4, 5)	9	H1
$\{1, 3, 5, 8\}$	4	1/5	6/13	4/13	(5, 8)	13	H1
$\{1, 4, 5, 6\}$	4	1/5	4/9	2/9	(4, 5)	9	H1
$\{1, 2, 4, 5, 6\}$	5	1/6	3/10	1/5	(4, 6)	10	H2
$\{1, 2, 4, 5, 6, 7\}$	6	1/7	4/11	2/11	(5, 6)	11	H2
$\{1, 2, 5, 6, 7, 8, 11\}$	7	1/8	3/13	2/13	(5, 8)	13	H2
<i>For comparison, the extremal set on which the bound is tight:</i>							
$\{1, 2, 3, 4, 5\}$	5	1/6	1/6	1/6	(1, 5)	6	none

The final row is instructive. On the arithmetic progression $\{1, \dots, n\}$, which is the instance Tao singles out as hardest, the envelope maximum equals the bound exactly and is witnessed by the speed-1 runner paired with v_n . The hypotheses were therefore well motivated by the extremal case. What the search shows is that the extremal case is not representative of the rest.

9 Discussion

9.1 Why the hypotheses failed

The motivating intuition was that the speed-1 runner is distinguished: its graph is a single triangle spanning $[0, 1]$, hence above $1/(n+1)$ except on a short initial and final segment. That is true, and it is precisely why the hypothesis fails.

A crossing is *valid* in the sense of Definition 5.6 only if it lies on the lower envelope, i.e. only if no runner is below it. A runner whose graph is high almost everywhere is, for that very reason, rarely the one attaining the minimum, and the envelope is built entirely out of the runners that do attain

it. Slow runners have wide triangles and spend most of $(0, 1)$ far from the origin. Fast runners have narrow triangles and return to the origin often, so they are the ones pressing the envelope down. On this reading the speed-1 runner is the least likely of all the runners to appear in a valid intersection, rather than the most.

The extremal set $\{1, \dots, n\}$ is the exception that generated the intuition in the first place. There every speed is small, no runner dominates the envelope, and the maximiser is realised by $(1, n)$ at $t = 1/(n + 1)$. A hypothesis calibrated on that case has little reason to hold once the speeds are allowed to spread out.

9.2 The witnessing pairs admit no simple description

By Corollary 5.3, $\kappa(V)$ is realised by a pair (v_i, v_j) and a modulus $m = v_i + v_j$. If the admissible pairs could be restricted, for instance to those containing v_n , then by Remark 5.5 the Haralambis maximum would need to be taken over far fewer moduli, which is exactly the kind of reduction exploited in [13] and [19]. We therefore tested seven natural restrictions exhaustively over all speed sets with $v_1 = 1$ and $v_n \leq 2n$.

These counts required a computation the other three search programs do not perform. `LRtable.py` and `LRtable2.py` ask only whether *some* valid lonely intersection has a given property, whereas the question here concerns the single intersection of greatest height. A fourth search program, `patterns.py`, was therefore written for this section. It shares the segment representation and exact arithmetic of the others, but by Corollary 5.3 it considers only rising–falling crossings and peaks, since no other breakpoint can realise the maximum.

Table 5: Seven candidate restrictions on the pair (v_i, v_j) realising $\kappa(V)$, tested over all speed sets with $v_1 = 1$, $v_n \leq 2n$, as computed by `patterns.py`. By Remark 5.4 a restriction is counted as satisfied when *some* pair realising $\kappa(V)$ meets it. Entries give the number of speed sets satisfying the restriction, out of 10, 35, 126, 462 and 1716 respectively.

Restriction on (v_i, v_j)	n					smallest failure
	3	4	5	6	7	
contains a speed $> n$	9	31	115	422	1 600	$\{1, 2, 3\}$
contains v_n	10	29	89	269	1 009	$\{1, 2, 3, 6\}$
contains v_1 or v_n	10	30	101	320	1 222	$\{1, 3, 4, 6\}$
$\gcd(v_i, v_j) = 1$	9	32	112	404	1 508	$\{1, 3, 6\}$
$v_i + v_j > n$	10	34	125	457	1 705	$\{1, 2, 3, 6\}$
adjacent in V	6	26	85	263	998	$\{1, 2, 3\}$
$\max(v_i, v_j) > n/2$	10	35	126	462	1 716	none for $n \leq 7$
(total speed sets)	10	35	126	462	1 716	

Six of the seven fail, and none of the six shows any tendency to improve as n grows. The witnesses are small and easy to check by hand. On $V = \{1, 2, 3\}$ the maximum $\kappa = 1/4$ is realised at $t = 1/4$ by the pair $(1, 3)$, the only maximising pair; it contains no speed exceeding $n = 3$ and is not adjacent in V . On $V = \{1, 3, 4, 6\}$ the maximum is realised by $(3, 4)$ alone, avoiding both v_1 and v_n .

The failure of the coprimality restriction deserves separate comment, since it is the one that would have connected most directly to existing work. On $V = \{1, 3, 6\}$ the maximum $\kappa = 1/3$ is realised at $t = 4/9$ by the pair $(3, 6)$ and by no other, and $\gcd(3, 6) = 3$. The pairings produced

by the coprime-matching argument of Bohman and Peng [5] are therefore not in general the pairs that realise κ , so the two notions of “the relevant pair of speeds” do not coincide and one cannot be substituted for the other without further argument.

The seventh restriction is the interesting one. Every speed set with $v_1 = 1$ and $v_n \leq 2n$ for $n \leq 7$ has a maximising pair whose larger member exceeds $n/2$, without exception across all 2349 such sets. Extending the search to $n = 8$ produces exactly one failure, namely

$$V = \{1, 2, 3, 4, 7, 8, 12, 13\}, \quad \kappa(V) = \frac{1}{5} \text{ at } t = \frac{1}{5},$$

whose unique maximising pair is $(1, 4)$ with $\max(v_i, v_j) = 4 = n/2$. The restriction therefore fails, but only just, and only by an equality. Whether the weaker statement $\max(v_i, v_j) \geq n/2$ survives is the most concrete question this data raises, and we have not tested it beyond $n = 8$.

The negative results are useful in their own right. On this evidence, a proof of the case $v_1 = 1$ that begins by singling out one distinguished speed, or one distinguished pair, is unlikely to work: the pairs realising κ range over essentially the whole speed set, and coprimality does not pick them out. What the seventh restriction suggests is that a *size* condition may be the right shape after all, since requiring only that the larger speed be comparable to n survives everything the other six do not.

9.3 Limitations

Several limitations should be kept in view.

The verification of Section 7.6 reaches $n \leq 10$ and the hypothesis searches reach $n \leq 7$ (H1) and $n \leq 9$ (H2); the cost of the pairwise crossing enumeration, not the enumeration of speed sets, is the binding constraint. A property verified over this range need not persist, and the results of Section 9.2 show concretely that patterns apparent in a handful of small cases can fail once the search is widened.

As noted in Remark 5.8, Definition 5.6 does not test peaks, and by Remark 5.7 it admits type-(ii) crossings that are not maximisers. Neither affects the conclusions in the range searched, and both were checked computationally, but a definitive implementation should handle them directly.

Finally, and most importantly, this report establishes no new case of Conjecture 1.1 and proves none of the observed patterns. Its results are the reduction of Section 4, the structural lemmas of Section 5, and a body of negative computational evidence that narrows the space of plausible geometric approaches.

10 Conclusion

Tao’s question of whether $v_n \leq 2n$ implies the Lonely Runner Conjecture reduces entirely to speed sets containing the speed 1, since for $v_1 \geq 2$ the classical bound $v_n \leq nv_1$ already applies. That reduction, recorded in Proposition 4.1, is what the rest of this report addresses.

Reading the distance functions as sawtooths turns $\kappa(V)$ into the height of the lower envelope of an arrangement of line segments. This gives an exact finite algorithm for computing κ , since the envelope is piecewise linear and its maximum lies at a peak or a crossing, and it also connects the picture back to the existing literature: a crossing of runners v_i and v_j sits at a time of the form $\ell/(v_i + v_j)$, which is precisely the modulus appearing in Haralambis’ characterisation of κ .

The software built on this framework reverified the conjecture across all 125 476 speed sets with $v_1 = 1$ and $v_n \leq 2n$ for $n \leq 10$, agreeing with the published verifications by an unrelated method. It then tested two geometric properties that would, if true, have given a route to the case $v_1 = 1$. Both

are false. The speed-1 runner need not witness the optimal lonely time, failing first at $V = \{1, 3, 4\}$; admitting the speed-2 runner as an alternative does not repair matters, failing at $V = \{1, 2, 4, 5, 6\}$. Of seven further descriptions of the pair of speeds realising κ , six fail outright, including coprimality; the seventh holds for every set with $n \leq 7$ and then fails at $n = 8$ by an equality.

The common thread is that no small, fixed feature of a speed set appears to be responsible for its lonely time. Slow runners sit high above the envelope precisely because they are slow, and so are the least likely to determine it; the pair that does determine it ranges over essentially the whole set. The arithmetic progression $\{1, \dots, n\}$, on which the bound is tight and on which both hypotheses hold, turns out to be a poor guide to the general case. A successful geometric argument for $v_1 = 1$ will therefore need to control the whole envelope rather than any distinguished part of it.

11 Directions for Further Work

The following are the directions the project would most naturally take if continued. They are recorded here in enough detail to be picked up independently of the present author.

- (1) Replace the pairwise crossing test by a sweep-line computation of the lower envelope, reducing the per-set cost from $O(n(\sum v_i)^2)$ to $O(\sum v_i \log n)$. This is the single change that would most extend the reachable range of n ; the hypothesis searches, not the verification, are what it would unblock.
- (2) Extend to the other search programs the orientation tagging already used by `patterns.py`, and add peak testing per Remark 5.8, so that every tool matches the theory of Section 5 exactly rather than coincidentally.
- (3) Export, for every valid lonely intersection, the pair (v_i, v_j) , the modulus $m = v_i + v_j$, $\gcd(v_i, v_j)$ and the height, to CSV, and search systematically for restrictions of the kind that Section 9.2 rules out in their simplest forms.
- (4) Extend the verification of Section 7.6 to $n = 11$ and $n = 12$, the largest values for which Conjecture 1.1 is known, and then beyond, where a failure would no longer be attributable to a bug.
- (5) Investigate whether the geometric picture gives any purchase on the coprime-matching approach of [5], in which the relevant pairings are also between speeds drawn from distinct intervals. The appearance of $\gcd(v_i, v_j)$ as a candidate invariant in item (3) suggests a possible point of contact.
- (6) Test the weakened size condition $\max(v_i, v_j) \geq n/2$ of Section 9.2 for $n \geq 9$. Of the seven restrictions tried it is the only one still standing, its strict form having failed at $n = 8$ by a single equality, and a size condition of this kind would restrict the Haralambis moduli in exactly the way Remark 5.5 requires.

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Use of artificial intelligence tools. Artificial intelligence tools were used in the preparation of this report in the following ways: to assist the author, an undergraduate student without graduate-level training in number theory, in understanding the research papers surveyed in Section 3; to assist in drafting and refining the exposition throughout; and to assist in reviewing and debugging the software described in Section 6 and Section 7. All mathematical statements attributed to other authors are cited to their original sources, and the author has verified the computational results reported here by direct execution of the programs. Responsibility for any errors remains the author's.

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A Reproducing the Results

All source code, together with the complete per-set output underlying Table 1, Table 2, Table 3, Table 4 and Table 5, is available at

<https://github.com/asfihassan/lonely-runner-geometric-search>

Each table in this report is reproduced by the program indicated below.

Table	Produced by
Table 1	<code>lrc.py</code> (or <code>timings.py</code> for the timings)
Table 2	<code>LRtable.py</code>
Table 3	<code>LRtable2.py</code>
Table 4	<code>LRCgraph.py</code> , per speed set
Table 5	<code>patterns.py</code>